

Crooked cylinder

X19 (2019) — English translation

A cylindrical vessel of height $H = 50$ cm with mirror walls is filled with a transparent liquid, the refractive index of which decreases with height according to a linear law. Near the bottom of the cylinder there is a point light source with power $P = 100$ W. The refractive index of the liquid near the bottom is $n_1 = 1.50$, and at the surface $n_2 = 1.44$. The bottom of the vessel and its lid are covered with a substance that absorbs all radiation that hits it

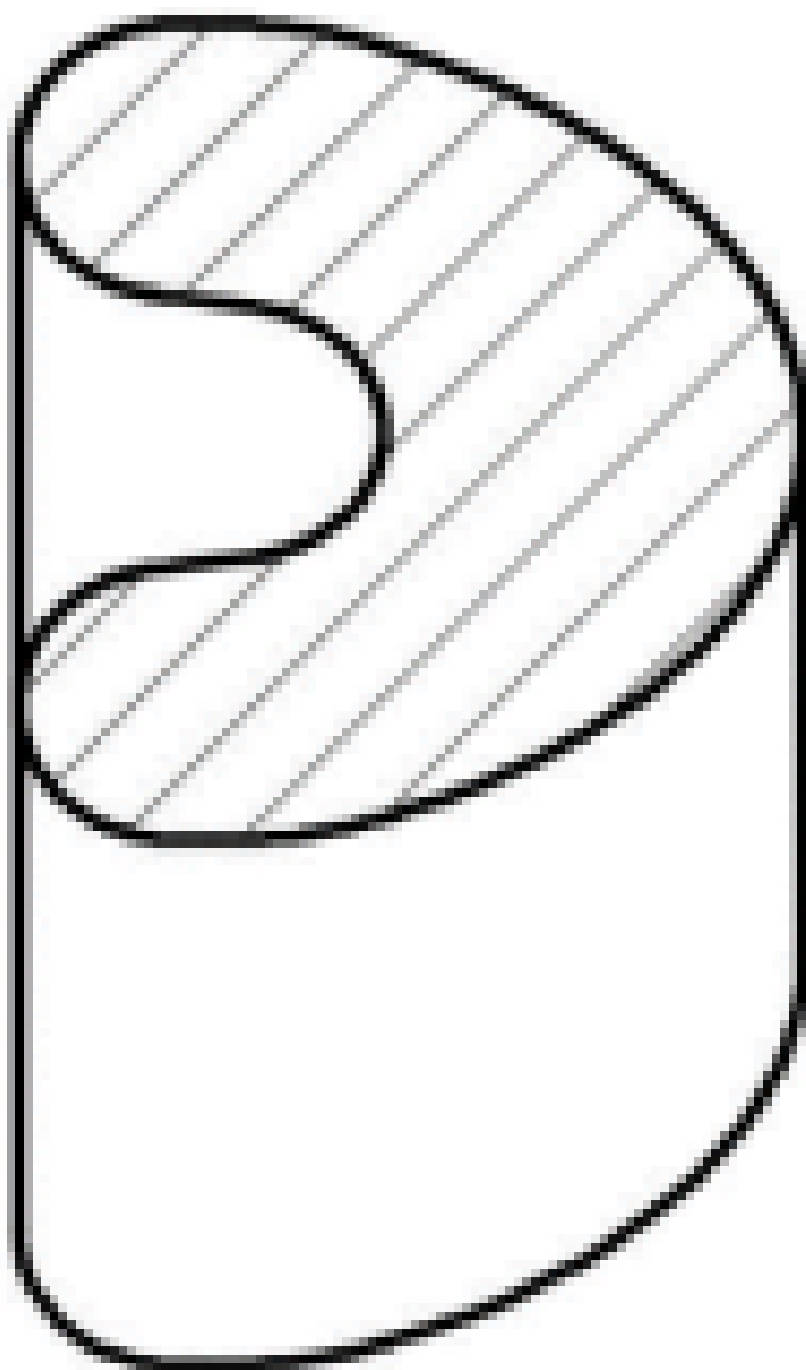


Figure 1: Figure 1.

A1	What radiation power P is absorbed by the bottom of the cylinder?	0
A2	Suppose a certain beam passes near the cylinder cover. What is the radius of curvature R of this ray?	0
A3	What is the radius of curvature R of the same ray directly near the source?	0

Source: <https://pho.rs/p/3059>